

SAT UZBEKISTAN

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Telegram @satuzbhunters

1.

$$\frac{x+1}{5x^2} = \frac{k}{x}$$

In the given equation, k is a constant. The solution to the given equation is $\frac{1}{174}$. What is the value of k ?

2.

$$\sqrt{k-x} = 52-x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

3.

$$-16(5x-3)^2 + 4(5x-4)^2$$

The given expression can be rewritten as

$$\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6},$$

where a , b , and c are constants. What is the value of $a+b+c$?

- A) -360 C) -60
B) -180 D) -10

4.

$$f(x) = 14x^2 - 35x + 7$$

The given equation defines the function f . For what values of x does $f(x)$ reach its minimum?

- A) 1.25 C) 7
B) 1.75 D) 35

5.

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants and $1 < a < 4$. The graph of $y = f(x)$ in the xy -plane passes through the points $(11, 0)$, and $(-5, 0)$. If a is an integer, what could be the value of $a + b$?

6.

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

- A) 81
- B) 129
- C) 192
- D) 225

7.

Which of the following expressions is (are) a factor of $4x^2 + 31x - 45$?

- I. $x - 9$
- II. $4x - 5$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

8.

$$f(x) = x + 4$$

$$g(x) = 3x^2 - rx + 48$$

The functions f and g are given. In function g , r is a constant. If $f(x) \cdot g(x) = 3x^3 + 192$, what is the value of r ?

9.

$$y = 2x^2 - 19x + 40$$

$$y = x + a$$

In the given system of equations, a is a constant. The graph of the equations in the given system intersect at exactly one point (x, y) , in the xy -plane. What is the value of x ?

- A) -10 C) 5
B) -5 D) 10

10.

$$\frac{1}{cx} = \frac{x}{152} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

11.

$$x^2 + 5x = 47$$

The solution to the given equation are $\frac{m + \sqrt{q}}{2}$ and $\frac{m - \sqrt{q}}{2}$, where m and q are integers. what is the value of $m + q$?

12.

$$r(t) = 55t - 2t^2$$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

- A) $1 + \left(\frac{55}{2}\right)^2$ C) $1 - 2\left(\frac{55}{4}\right)^2$
B) $1 + 2\left(\frac{55}{4}\right)^2$ D) $1 - \left(\frac{55}{2}\right)^2$

13.

$$3w^2 + 11w - c = 0$$

In the given equation, c is a constant. A solution to the given equation is $\frac{-11+\sqrt{433}}{6}$. What is the value of c ?

14.

$$x^2 - 2x - 18 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

- A) 17
- B) 19
- C) 38
- D) 76

15.

The function f is defined by

$$f(x) = ax^2 + bx + c,$$

where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(10, 0)$ and $(-5, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- | | |
|-------|------|
| A) -8 | A) 5 |
| A) -4 | A) 6 |

16.

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,600 feet 10 seconds after it was launched. Based on the model, what was the height, in feet, of the object 7 seconds after it was launched?

17.

$$(x - k)^2 = (k - 4a)(x - k)$$

In the given equation, a and k are constants, where $k > 4a$. The sum of the solutions to the equation is $3k + 35$. What is the value of a ?

18.

The function $f(x) = \frac{1}{9}(x - 6)^2 + 3$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy plane?

- A) The toy car's minimum height was 3 inches above the ground.
- B) The toy car's minimum height was 6 inches above the ground.
- C) The toy car's height was 3 inches above the ground when it started moving.
- D) The toy car's height was 6 inches above the ground when it started moving.

19.

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?



20.

$$f(x) = 5x^2 + 60x + 181$$

The function g is defined by $g(x) = f(x+8)$. What is the minimum value of $g(x)$?

A) -14 C) 1 B) -6 D) 9

21.

In the xy -plane, a parabola has a vertex $(7, 12)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

A) 10 B) 14 C) 18 D) 36

22.

If the vertex of a parabola is at the point $(1, 24)$ and the quadratic can be written in the form $-a(x^2) + bx + c$ and a , b , and c are all positive integers, what is the maximum value of b ?

A) 46 B) 23 C) -46 D) -23

23.

In the xy -plane, a parabola has a vertex $(5,1)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = -ax^2 + bx + c$, where a, b , and c are constants and $a > 0$, which of the following could be a value of c ?

- A) -5
- B) 2
- C) 3
- D) 9

24.

The given quadratic function $54x^4 + 219x^2 + 105$ has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d , and k are integers, what is the smallest possible value of ab ?

25.

The function f is defined by $f(x) = ax^2 + bx + c$, where a, b and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(13, 0)$ and $(-6, 0)$. Which of the following is the value of $a + b$ in terms of a ?

- A) $7a$
- B) $6a$
- C) $-6a$
- D) $-7a$

26.

Which of the following has $x + 2a$ as a factor where a is a positive integer?

- A) $3x^2 + 7x + 14a = 0$
- B) $3x^2 + 28x + 14a = 0$
- C) $3x^2 + 42x + 14a = 0$
- D) $3x^2 + 49x + 14a = 0$

27.

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(-7, 14)$ and passes through the point $(-4, -31)$ and has a y -intercept at $(0, a)$. The graph of $3f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

28.

$$f(x) = (x + a)(x + b)$$

For the given function $f(x)$, where a and b are integer constants, $f(-21) > 0$, $f(-15) > 0$, and $f(-18) < 0$, what is one possible value of $a + b$?

29.

The vertex of the function $f(x)$ is given as $(9, -13)$, and the point $(3, -4)$ lies on the parabola. If $g(x) = 4f(x - 2)$, what is the value of $g(0) - f(0)$?

30.

- For the given parabola $ax^2 + bx + c$, a , b , and c are constants. The x -intercepts of the parabola are at $(14, 0)$ and $(k, 0)$. If $f(27) = f(11)$, what is the value of k ?

31.

$$f(x) = x^2 + 5x - 6$$
$$g(x) = x^2 + 2x - 35$$

The quadratic function $f(x)$ has 2 solutions j and k , where $j < k$. The quadratic function $g(x)$ has 2 solutions l and m , where $l < m$. The quadratic function $h(x) = x^2 + 7x + c$ has solutions of $k + l$ and $m + j$, and can be rewritten as $(x + a)(x + b)$, what is the value of c ?

32.

The graph of the quadratic function $y = f(x)$ in the xy -plane intersects the x -axis when $x = 74$ and $x = k$, where k is a constant. The maximum value of $y = f(x)$ occurs at the point $(17, m)$, where m is a constant. What is the value of k ?

33.

x	$f(x)$
-9	155
-3	227
3	155

Three points on the quadratic function $f(x)$ are given in the table. If $g(x) = f(x + 4)$, what is the y-intercept of the function $g(x)$?

34.

$$f(x) = (x-a)(x-b)$$

The function f is defined above, where a and b are integer constants. If $f(1) > 0$, $f(4) < 0$, and $f(7) > 0$, what is one possible value of $a+b$?

35.

$$7rx^2 - 28sx + 7r$$

In the given equation, r is a positive constant, and s is a negative constant. The equation has two solutions. If $\frac{r}{s} > k$, what's the minimum value of k ?

36.

$$34z^{14} + bz^7 + 70$$

In the given expression, b is a positive integer. If $qz^7 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

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37.

$$9x^2 + 30x + q = 43$$

In the given equation, q is a constant. One solution to the given equation is $\frac{-5+\sqrt{14}}{3}$. What is the value of q ?

A) 54

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B) 43

C) 11

D) 5

38.

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$. The rational function g is defined by $g(x) = \frac{f(x)}{x-17}$. In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

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39.

For the function f , for each increase in the value of x by c , where c is a positive constant, the value of $f(x)$ increases by a factor of

27. Which of the following equivalent forms of the function f displays $\frac{1}{c}$ as a coefficient of x ?

(A) $f(x) = 73(3)^{\frac{1}{2}x}$

(B) $f(x) = 73(3^3)^{\frac{1}{6}x}$

(C) $f(x) = 73(9)^{\frac{1}{4}x}$

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(D) $f(x) = 73(27^{\frac{1}{3}x})^{\frac{1}{2}}$

40.

$g(x) = x^3 + ax^2 + bx + c$ @satuzbhunters

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

41.

$f(x) = x^2 - 4x - 320$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

(A) $f(x) = x^2 - 4(x + 80)$

(B) $f(x) = (x - 2)^2 + (-324)$

(C) $f(x) = x(x - 4) + (-320)$

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(D) $f(x) = (x + 16)(x - 20)$

42.

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m ?

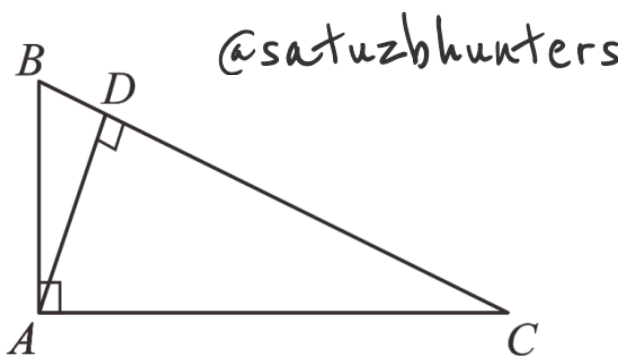
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43.

$$f(x) = -0.0038400x^2 + 15.236x - 15,103$$

The model shown gives the predicted average Arctic sea ice area $f(x)$, in millions of square kilometers, for September of each year x , where x represents a year and $1979 \leq x \leq 2012$. Based on the model, what is the positive difference, in millions of square kilometers, between the predicted average Arctic sea ice area for September of the year 1991 and the predicted average Arctic sea ice area for September of the year 1992? (Round your answer to the nearest thousandth.)

44.



In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . The measure of $\angle ABD$ is w° , the measure of $\angle CAD$ is z° , and $AD = 178$. If $\cos w^\circ = \frac{9}{41}$, what is the value of $\tan z^\circ$?

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